A bizarre example of what appears to be a Tit for Tat arrangement in nature was discovered by Eric Fischer in a hermaphrodite fish, the sea bass. Unlike us, these fish don't have their sex determined at conception by their chromosomes. Instead, every individual is capable of performing both female and male functions. In any one spawning episode they shed either eggs or sperm. They form monogamous pairs and, within the pair, take turns to play the male and female roles. Now, we may surmise that any individual fish, if it could get away with it, would 'prefer' to play the male role all the time, because the male role is cheaper. Putting it another way, an individual that succeeded in persuading its partner to play the female most of the time would gain all the benefits of 'her' economic investment in eggs, while 'he' has resources left over to spend on other things, for instance on mating with other fish.

In fact, what Fischer observed was that the fishes operate a system of pretty strict alternation. This is just what we should expect if they are playing Tit for Tat. And it is plausible that they should, because it does appear that the game is a true Prisoner's Dilemma, albeit a somewhat complicated one. To play the COOPERATE card means to play the female role when it is your turn to do so. Attempting to play the male role when it is your turn to play the female is equivalent to playing the DEFECT card. Defection is vulnerable to retaliation: the partner can refuse to play the female role next time it is 'her' (his?) turn to do so, or 'she' can simply terminate the whole relationship. Fischer did indeed observe that pairs with an uneven sharing of sex roles tended to break up.

A question that sociologists and psychologists sometimes ask is why blood donors (in countries, such as Britain, where they are not paid) give blood. I find it hard to believe that the answer lies in reciprocity or disguised selfishness in any simple sense. It is not as though regular blood donors receive preferential treatment when they come to need a transfusion. They are not even issued with little gold stars to wear. Maybe I am naïve, but I find myself tempted to see it as a genuine case of pure, disinterested altruism. Be that as it may, blood-sharing in vampire bats seems to fit the Axelrod model well. We learn this from the work of G. S. Wilkinson.

Vampires, as is well known, feed on blood at night. It is not easy for them to get a meal, but if they do it is likely to be a big one. When dawn comes, some individuals will have been unlucky and return completely empty, while those individuals that have managed to find a victim are